

ACR13943

## 1,1,2-Trichloroethane

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**产品说明:** 1,1,2-三氯乙烷  
**Product Description:** 1,1,2-Trichloroethane

**Cat No. :** 139430000; 139430010; 139432500  
**Synonyms:** beta-Trichloroethane; Ethane trichloride; Vinyl trichloride  
**CAS No** 79-00-5  
**Molecular Formula** C2 H3 Cl3

**Supplier**

**UK entity/business name**  
Fisher Scientific UK  
Bishop Meadow Road,  
Loughborough, Leicestershire LE11 5RG, United Kingdom

**EU entity/business name**  
Thermo Fisher Scientific  
Janssen Pharmaceuticaaan 3a, 2440 Geel, Belgium

**Emergency Telephone Number** For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

**E-mail address** begel.sdsdesk@thermofisher.com

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

### SECTION 2. HAZARD IDENTIFICATION

**Physical State**  
Liquid

**Appearance**  
Clear

**Odor**  
sweet

#### Emergency Overview

Toxic if inhaled. Harmful to aquatic life with long lasting effects. Harmful if swallowed. Harmful in contact with skin. Suspected of causing cancer. Repeated exposure may cause skin dryness or cracking.

#### Classification of the substance or mixture

|                                    |            |
|------------------------------------|------------|
| Acute Oral Toxicity                | Category 4 |
| Acute Dermal Toxicity              | Category 4 |
| Acute Inhalation Toxicity - Vapors | Category 3 |
| Carcinogenicity                    | Category 2 |
| Chronic aquatic toxicity           | Category 3 |

#### Label Elements

## 1,1,2-Trichloroethane



## Signal Word

Danger

## Hazard Statements

H331 - Toxic if inhaled

H412 - Harmful to aquatic life with long lasting effects

H351 - Suspected of causing cancer

H302 + H312 - Harmful if swallowed or in contact with skin

## Precautionary Statements

## Prevention

P201 - Obtain special instructions before use

P202 - Do not handle until all safety precautions have been read and understood

P261 - Avoid breathing dust/fume/gas/mist/vapors/spray

P264 - Wash face, hands and any exposed skin thoroughly after handling

P270 - Do not eat, drink or smoke when using this product

P271 - Use only outdoors or in a well-ventilated area

P280 - Wear protective gloves/protective clothing/eye protection/face protection

## Response

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P311 - Call a POISON CENTER or doctor

P330 - Rinse mouth

P362 + P364 - Take off contaminated clothing and wash it before reuse

## Storage

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

P405 - Store locked up

## Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Physical and Chemical Hazards

None identified.

## Health Hazards

Toxic if inhaled. Harmful if inhaled. Harmful if swallowed. Harmful in contact with skin. Suspected of causing cancer.

## Environmental hazards

Harmful to aquatic life with long lasting effects. . Will likely be mobile in the environment due to its water solubility. The product is water soluble, and may spread in water systems.

## Other Hazards

Toxic to terrestrial vertebrates. This product does not contain any known or suspected endocrine disruptors.

## SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

| Component             | CAS No  | Weight % |
|-----------------------|---------|----------|
| 1,1,2-Trichloroethane | 79-00-5 | <=100    |

## SECTION 4. FIRST AID MEASURES

## General Advice

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

## Eye Contact

## 1,1,2-Trichloroethane

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.

**Ingestion**

Do NOT induce vomiting. Call a physician or poison control center immediately.

**Most important symptoms and effects**

None reasonably foreseeable.

**Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Vapors may form explosive mixtures with air. Non-combustible. Thermal decomposition can lead to release of irritating gases and vapors. Keep product and empty container away from heat and sources of ignition.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Ensure adequate ventilation. Use personal protective equipment as required. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas.

**Environmental Precautions**

Should not be released into the environment. Do not flush into surface water or sanitary sewer system.

**Methods for Containment and Clean Up**

Keep in suitable, closed containers for disposal. Soak up with inert absorbent material.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

## 1,1,2-Trichloroethane

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance.

**Storage**

Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Keep away from heat/sparks/open flames/hot surfaces.  
- No smoking.

**Specific Use(s)**

Use in laboratories

## SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

**Control Parameters**

| Component             | China | Taiwan                                   | Thailand    | Hong Kong                                |
|-----------------------|-------|------------------------------------------|-------------|------------------------------------------|
| 1,1,2-Trichloroethane | -     | TWA: 10 ppm<br>TWA: 55 mg/m <sup>3</sup> | TWA: 10 ppm | TWA: 10 ppm<br>TWA: 55 mg/m <sup>3</sup> |

| Component             | ACGIH TLV           | OSHA PEL                                                                                                         | NIOSH                                                     | The United Kingdom | European Union |
|-----------------------|---------------------|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|--------------------|----------------|
| 1,1,2-Trichloroethane | TWA: 10 ppm<br>Skin | (Vacated) TWA: 10 ppm<br>(Vacated) TWA: 45 mg/m <sup>3</sup><br>Skin<br>TWA: 10 ppm<br>TWA: 45 mg/m <sup>3</sup> | IDLH: 100 ppm<br>TWA: 10 ppm<br>TWA: 45 mg/m <sup>3</sup> | -                  |                |

Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Exposure Controls****Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

**Personal protective equipment**

**Eye Protection** Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Nitrile rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Neoprene       | recommendations   |                 |             |                       |
| Natural rubber |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.

(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Wear appropriate protective gloves and clothing to prevent skin exposure

**Respiratory Protection** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

## 1,1,2-Trichloroethane

|                                        |                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly                                                                                                                                                                                                         |
| <b>Large scale/emergency use</b>       | Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced<br><b>Recommended Filter type:</b> Organic gases and vapours filter Type A Brown conforming to EN14387                                                                  |
| <b>Small scale/Laboratory use</b>      | Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.<br><b>Recommended half mask:-</b> Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141<br>When RPE is used a face piece Fit Test should be conducted |
| <b>Hygiene Measures</b>                | Handle in accordance with good industrial hygiene and safety practice.                                                                                                                                                                                                                                                      |
| <b>Environmental exposure controls</b> | Prevent product from entering drains.                                                                                                                                                                                                                                                                                       |

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                                |                                     |                                          |
|------------------------------------------------|-------------------------------------|------------------------------------------|
| <b>Appearance</b>                              | Clear                               |                                          |
| <b>Physical State</b>                          | Liquid                              |                                          |
| <b>Odor</b>                                    | sweet                               |                                          |
| <b>Odor Threshold</b>                          | No data available                   |                                          |
| <b>pH</b>                                      | No information available            |                                          |
| <b>Melting Point/Range</b>                     | -37 °C / -34.6 °F                   |                                          |
| <b>Softening Point</b>                         | No data available                   |                                          |
| <b>Boiling Point/Range</b>                     | 110 - 115 °C / 230 - 239 °F         | @ 760 mmHg                               |
| <b>Flash Point</b>                             | No information available            | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>                        | No data available                   |                                          |
| <b>Flammability (solid,gas)</b>                | Not applicable                      | Liquid                                   |
| <b>Explosion Limits</b>                        | <b>Lower</b> 6<br><b>Upper</b> 15.5 |                                          |
| <b>Vapor Pressure</b>                          | 20 @ 25 mbar °C                     |                                          |
| <b>Vapor Density</b>                           | 4.63 (Air = 1.0)                    | (Air = 1.0)                              |
| <b>Specific Gravity / Density</b>              | 1.430                               |                                          |
| <b>Bulk Density</b>                            | Not applicable                      | Liquid                                   |
| <b>Water Solubility</b>                        | 4 g/L (20°C)                        |                                          |
| <b>Solubility in other solvents</b>            | No information available            |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                                     |                                          |
| <b>Component</b>                               | <b>log Pow</b>                      |                                          |
| 1,1,2-Trichloroethane                          | 1.89                                |                                          |
| <b>Autoignition Temperature</b>                | 459 °C / 858.2 °F                   |                                          |
| <b>Decomposition Temperature</b>               | No data available                   |                                          |
| <b>Viscosity</b>                               | 1.69 cP at 25 °C                    |                                          |
| <b>Explosive Properties</b>                    | No information available            |                                          |
| <b>Oxidizing Properties</b>                    | No information available            |                                          |
| <b>Molecular Formula</b>                       | C2 H3 Cl3                           |                                          |
| <b>Molecular Weight</b>                        | 133.4                               |                                          |

## SECTION 10. STABILITY AND REACTIVITY

|                                 |                                          |
|---------------------------------|------------------------------------------|
| <b>Stability</b>                | Stable under normal conditions.          |
| <b>Hazardous Reactions</b>      | None under normal processing.            |
| <b>Hazardous Polymerization</b> | Hazardous polymerization does not occur. |

**SAFETY DATA SHEET****1,1,2-Trichloroethane****Conditions to Avoid** Excess heat. Incompatible products.**Materials to avoid** Bases. Strong oxidizing agents. Strong bases. Metals.**Hazardous Decomposition Products** Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Chlorine. Phosgene. Hydrogen chloride gas.**SECTION 11. TOXICOLOGICAL INFORMATION****Product Information****(a) acute toxicity;**

| Component             | LD50 Oral                | LD50 Dermal                  | LC50 Inhalation              |
|-----------------------|--------------------------|------------------------------|------------------------------|
| 1,1,2-Trichloroethane | LD50 = 836 mg/kg ( Rat ) | LD50 = 5371 mg/kg ( Rabbit ) | LC50 = 2.78 mg/L ( Rat ) 8 h |

**(b) skin corrosion/irritation;** No data available**(c) serious eye damage/irritation;** No data available**(d) respiratory or skin sensitization;**

|                    |                   |
|--------------------|-------------------|
| <b>Respiratory</b> | No data available |
| <b>Skin</b>        | No data available |

**(e) germ cell mutagenicity;** No data available**(f) carcinogenicity;** Category 2

The table below indicates whether each agency has listed any ingredient as a carcinogen

**(g) reproductive toxicity;** No data available**(h) STOT-single exposure;** No data available**(i) STOT-repeated exposure;** No data available**Target Organs** None known.**(j) aspiration hazard;** No data available**Symptoms / effects, both acute and delayed** No information available**SECTION 12. ECOLOGICAL INFORMATION****Ecotoxicity effects**

Do not empty into drains. Contains a substance which is: Harmful to aquatic organisms. The product contains following substances which are hazardous for the environment. Harmful to aquatic organisms, may cause long-term adverse effects in the aquatic environment.

| Component             | Freshwater Fish                                           | Water Flea                                      | Freshwater Algae                                       | Microtox              |
|-----------------------|-----------------------------------------------------------|-------------------------------------------------|--------------------------------------------------------|-----------------------|
| 1,1,2-Trichloroethane | LC50: = 81.6 mg/L, 96h flow-through (Pimephales promelas) | EC50: 57 - 110 mg/L, 48h Static (Daphnia magna) | EC50: = 167 mg/L, 96h static (Desmodesmus subspicatus) | EC50 = 105 mg/L 5 min |

**SAFETY DATA SHEET****1,1,2-Trichloroethane**

|  |                                                            |                                         |  |  |
|--|------------------------------------------------------------|-----------------------------------------|--|--|
|  | LC50: 35 - 47 mg/L,<br>96h static (Lepomis<br>macrochirus) | EC50: = 18 mg/L, 48h<br>(Daphnia magna) |  |  |
|--|------------------------------------------------------------|-----------------------------------------|--|--|

**Persistence and Degradability****Persistence**

Persistence is unlikely.

**Degradation in sewage  
treatment plant**

Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**Bioaccumulative Potential**

Bioaccumulation is unlikely

| Component             | log Pow | Bioconcentration factor (BCF) |
|-----------------------|---------|-------------------------------|
| 1,1,2-Trichloroethane | 1.89    | 0.7 - 6.7 dimensionless       |

**Mobility in soil**

The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility Highly mobile in soils

**Endocrine Disruptor Information**

This product does not contain any known or suspected endocrine disruptors

**Persistent Organic Pollutant**

This product does not contain any known or suspected substance

**Ozone Depletion Potential**

This product does not contain any known or suspected substance

**SECTION 13. DISPOSAL CONSIDERATIONS****Waste from Residues/Unused  
Products**

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point.

**Other Information**

Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not let this chemical enter the environment.

**SECTION 14. TRANSPORT INFORMATION****Road and Rail Transport**

|                         |                               |
|-------------------------|-------------------------------|
| UN-No                   | UN2810                        |
| Proper Shipping Name    | Toxic liquid, organic, n.o.s. |
| Technical Shipping Name | (1,1,2-TRICHLOROETHANE)       |
| Hazard Class            | 6.1                           |
| Packing Group           | III                           |

**IMDG/IMO**

|                         |                               |
|-------------------------|-------------------------------|
| UN-No                   | UN2810                        |
| Proper Shipping Name    | Toxic liquid, organic, n.o.s. |
| Technical Shipping Name | (1,1,2-TRICHLOROETHANE)       |
| Hazard Class            | 6.1                           |
| Packing Group           | III                           |

**IATA**

|                         |                               |
|-------------------------|-------------------------------|
| UN-No                   | UN2810                        |
| Proper Shipping Name    | Toxic liquid, organic, n.o.s. |
| Technical Shipping Name | (1,1,2-TRICHLOROETHANE)       |
| Hazard Class            | 6.1                           |

## 1,1,2-Trichloroethane

Packing Group III

Special Precautions for User No special precautions required

## SECTION 15. REGULATORY INFORMATION

## International Inventories

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component             | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|-----------------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| 1,1,2-Trichloroethane | X                                                   | -                                       | X    | X     | 201-166-9 | X    | X   | X     | X    | X    | X    | KE-34069 |

## National Regulations

## SECTION 16. OTHER INFORMATION

Creation Date 01-May-2012  
Revision Date 05-Apr-2024  
Revision Summary SDS sections updated, 2, 3.

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Legend

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit

ACGIH - American Conference of Governmental Industrial Hygienists

DNEL - Derived No Effect Level

RPE - Respiratory Protective Equipment

LC50 - Lethal Concentration 50%

NOEC - No Observed Effect Concentration

PBT - Persistent, Bioaccumulative, Toxic

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer

PNEC - Predicted No Effect Concentration

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

POW - Partition coefficient Octanol:Water

vPvB - very Persistent, very Bioaccumulative

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road

OECD - Organisation for Economic Co-operation and Development

BCF - Bioconcentration factor

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

MARPOL - International Convention for the Prevention of Pollution from Ships

ATE - Acute Toxicity Estimate

VOC - (Volatile Organic Compound)



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**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**